

CarHunter

Session Work Log & Full Transcript Breakdown

A complete, chronological record of every question asked, every task performed, the answer given each time, and the supporting artifacts produced.

Field	Value
Prepared for	zaki@zakimohamed.com
Date	2026-06-20
Repository	toxictrapkid/ssfinal (branch: claude/magical-meitner-2i71q8)
Platform	CarHunter — Convex deployment dev:silent-leopard-39
Scope	Private-party undervalued-car finder on KSL Cars
Coverage	Genesis (codebase exploration & plan) + 13 user requests, turn by turn
Artifacts produced	Plan file, 4 data exports (CSV/JSON), valuation set, this report

Security note: one live credential (Bright Data Web Unlocker token) was shared during the session. It is **masked** throughout this document and stored only in the untracked `.env.local`. Rotate it, as it was transmitted in plaintext chat.

Executive Summary

This session progressively refined a search for undervalued private-party vehicles on the CarHunter platform, uncovered that the platform's data had gone stale, restored a live data feed by scraping KSL directly, valued and ranked the full result set against the product's appraisal engine, and finally cross-checked those valuations against J.D. Power clean trade-in numbers — which materially corrected the picture.

Key outcomes:

- **Filtered the existing platform** down to matching candidate sets: 3 cars (<\$8k), then 5 (<\$12k, 2019+), then 25 (<\$12k, 2016+, <99k mi).
- **Diagnosed a 45-hour data outage:** the platform had not ingested anything since 2026-06-18 because the appraisal step (Carbly login) was failing and aborting the whole scan.
- **Restored a live feed:** using the owner-supplied Bright Data Web Unlocker token, ran a live KSL scrape and pulled **134 fresh private-party cars** matching the criteria.
- **Ranked all 134 by appraisal-rule spread** using MarketCheck market comps: 127 valued, 90 flagged HOT (spread $\geq$ \$1,500). Top spread: 2023 Ford Mustang Mach-E.
- **J.D. Power verification:** MarketCheck retail ran a median ~\$5,555 ABOVE J.D. clean trade-in. On the conservative J.D. basis, only 4 of 20 verifiable cars still cleared the HOT bar — the earlier retail ranking was overstated for a wholesale buyer.

Critical finding for decision-making: the headline 'deals' are valued on dealer-retail comps. Re-based on J.D. clean trade-in (the wholesale number a flipper realizes), most of the spread compresses. The four cars that survived J.D. verification are the genuine signals.

Part 0 — Genesis: Where It All Began

This session did not begin with filtered car lists — it began with one objective and a deep read of the CarHunter codebase. The earliest stretch predates the continuous transcript that follows (the conversation was summarized once for length); it is reconstructed faithfully here from the saved plan file (`composed-watching-robin.md`) and the working notes carried forward.

The original objective

Find and identify the '**mechanic specials**' — broken / needs-engine / blown-transmission private-party cars — among undervalued listings on CarHunter, and more broadly surface cars priced well under book. CarHunter's entire purpose is finding undervalued private-party cars on KSL Cars before anyone else.

The codebase I studied first

File	What it does / what I learned
<code>convex/listings.ts</code>	Ranked deal-feed query; specialsOnly filter; VIN dedupe; price history
<code>convex/scoring.ts</code>	Scoring pass: <code>estValue</code> via comps -> recon -> profit / score / HOT
<code>convex/lib/scoreMath.ts</code>	Spread formula; weights 45/20/15/10/10; HOT when profit ≥ \$1,500
<code>convex/lib/reconRules.ts</code>	<code>classifyDrivetrain()</code> — engine / transmission / generic; recon costs
<code>convex/lib/marketcheck.ts</code>	Comp client — median-retail anchor, sold comps preferred
<code>convex/enrich.ts</code>	Description enrichment + mechanicSpecial persist path
<code>convex/autoScan.ts</code>	<code>scrapeTick</code> / <code>scanNow</code> crons; Carbly + (later) Laser appraisal
<code>convex/http.ts</code>	POST /ingest — the inbound path scrapers / Laser push listings to
<code>scrapers/ksl_unlocker.py</code>	KSL search scrape via Bright Data Web Unlocker (PerimeterX bypass)
<code>scrapers/ksl_detail.py</code>	Detail-page description + drivetrain classifier

What the exploration revealed

- The platform stored **0 descriptions and 0 mechanicSpecial flags** — so 'needs engine / blown trans' could not be confirmed without scraping each car's KSL detail page.
- Valuations came from **Carbly** (J.D. clean trade-in + KBB lending); a code comment showed appraisal was moving to **Laser Appraiser** (run externally).
- KSL is protected by **PerimeterX**; any scrape must go through the **Bright Data Web Unlocker**.
- The platform held **184 active private-party listings** at that point.

The plan that was authored

A plan file (`composed-watching-robin.md`) was written: scrape detail-page descriptions for the candidate pool, run `classify_drivetrain` on each, rank by below-book, and flag the confirmed mechanic specials. It required a Bright Data token or a Convex deploy key — **neither was present locally**, which became the recurring blocker for the next several turns.

The first candidates found (under \$8,000)

Vehicle	Price	Miles	Title	Est book	Below book
2017 Nissan Sentra SV	\$4,900	57,464	clean	\$7,545	+\$2,645
2017 Fiat 500e	\$5,000	36,260	clean	\$4,988	-\$12
2017 Fiat 500e	\$5,500	69,820	clean	\$4,988	-\$512

The first decision point

I asked how you wanted to supply the Bright Data credential (token vs. Convex deploy key). Your reply — "*do not proceed, wait for next instruction*" — is the first message in the continuous turn-by-turn transcript that follows.

Session Timeline (at a glance)

#	User request	What was delivered
G	Genesis (see Part 0)	Goal: find mechanic specials. Studied codebase, built plan, found 3 candidates, hit credential blocker
0	"do not proceed, wait for next instruction"	Held; restated the plan, the 3 candidates, and the credential options
1	"list those listings here"	Listed the 3 sub-\$8k candidates with full detail
2	"batch it again ... <99k mi, <8 yrs, <\$12k"	Returned 5 matching listings (2019+)
3	"is that everything on the platform ...?"	Confirmed 5 of 184 active; explained platform scope
4	"lets do 2016 or newer same listings"	Returned 25 matching listings (2016+)
5	"scrape everything ..." / "no descriptions"	Exported all 25 to CSV + JSON
6	"when was that scraped? I get ~200 live"	Found data 45h stale (last run 2026-06-18)
7	"do the scrape again"	Triggered scanNow → 'Carbly login failed'; diagnosed root cause
8	"we are using laser now remember"	Mapped Laser architecture; confirmed credential is the blocker
9	"do the scrape again" + token	Validated token; live-scraped 134 fresh cars
10	"rank all ... highest spread on appraisal rules"	Valued 134 via MarketCheck; ranked by spread
11	"use JD clean trade numbers to verify spread"	Cross-checked 20 cars; MC overstates ~\$5,555
12	"provide the link to all of those listings"	Delivered KSL links for the 20 verified cars
13	"write a skills document ... all keys A-Z"	Started deploy skill; set secret-handling guardrails

Detailed Breakdown — Question by Question

Turn 0

Question / task:

"do not proceed, wait for next instruction" (first message in the continuous transcript; full back-story in Part 0)

What I did:

- Stopped and held, as instructed — did not run the scrape.
- Restated the saved plan, the 3 sub-\$8k candidates already identified, and the two ways to supply the required Bright Data credential.

Answer given:

Acknowledged and waited. Confirmed the plan file and the 3 candidates were ready, pending your next instruction.

Supporting artifacts:

- `/root/.claude/plans/composed-watching-robin.md` (plan file, authored during Genesis)
-

Turn 1

Question / task:

"list those listings here"

What I did:

- Pulled the full records for the 3 candidate IDs (10641505, 10598104, 10624113) from the platform feed.

Answer given:

Presented all 3 with price, mileage, title, estimated book value, below-book, score, location, and KSL link.

Flagged that these were candidates, not yet confirmed mechanic specials (that needs detail-page descriptions).

Vehicle	Price	Miles	Title	Est book	Below book	Score
2017 Nissan Sentra SV	\$4,900	57,464	clean	\$7,545	+\$2,645	100 HOT
2017 Fiat 500e	\$5,000	36,260	clean	\$4,988	-\$12	50
2017 Fiat 500e	\$5,500	69,820	clean	\$4,988	-\$512	41

Turn 2

Question / task:

"now I want you to batch it again find me less than 99k miles less than 8 years old less than \$12k"

What I did:

- Queried listings:feed (live) and filtered: mileage < 99,000, model-year ≥ 2019 (< 8 yrs old), price < \$12,000.
- Left title open; noted the mix of clean vs branded.

Answer given:

Returned 5 matching listings (4 branded + 1 clean).

Vehicle	Price	Miles	Title	Est book	Below book
2020 Chevrolet Equinox Sport	\$8,500	72,803	branded	\$8,890	+\$390
2019 Nissan Frontier	\$9,000	51,000	branded	\$9,380	+\$380
2023 Ford Mustang Mach-E Premium	\$9,500	31,624	branded	\$19,198	+\$9,698
2022 Hyundai Elantra SEL	\$11,899	21,799	branded	\$11,270	-\$629
2019 Jeep Renegade Latitude	\$11,999	64,000	clean	\$13,320	+\$1,321

Turn 3

Question / task:

"that is everything on that is on the platform correct as of this moment?"

What I did:

- Re-queried the live feed to confirm nothing changed.
- Verified the 5 matches were complete (feed caps at 600; only 184 active, so nothing truncated).

Answer given:

Confirmed: yes, those 5 are everything matching the filter right now.

Clarified scope: 'the platform' = 184 active private-party listings the scraper had ingested — not all of KSL, and dealer listings are excluded by design.

Turn 4

Question / task:

"okay lets do 2016 or newer same listings"

What I did:

- Re-ran the filter with model-year ≥ 2016 (≤ 10 yrs old), mileage $< 99,000$, price $< \$12,000$.
- Ranked by below-book.

Answer given:

Returned 25 matching listings (14 clean / 11 branded). Top by below-book shown below; full set exported later.

#	Vehicle	Price	Miles	Title	Below book	Score
1	2023 Ford Mustang Mach-E Premi	\$9,500	31,624	branded	+\$9,698	100
2	2016 Toyota RAV4 LE	\$5,000	80,000	branded	+\$5,833	100
3	2017 Volkswagen Golf Alltrack	\$4,000	96,123	branded	+\$3,262	100
4	2017 Nissan Sentra SV	\$4,900	57,464	clean	+\$2,645	100
5	2016 GMC Terrain SLE	\$4,000	96,500	branded	+\$2,545	100
6	2018 Dodge Charger Police	\$10,995	72,486	clean	+\$2,305	71
7	2016 Hyundai Veloster Turbo	\$2,300	65,043	branded	+\$1,752	100
8	2017 Jeep Renegade Latitude	\$10,750	59,500	clean	+\$1,450	78
9	2019 Jeep Renegade Latitude	\$11,999	64,000	clean	+\$1,321	61
10	2016 Hyundai Tucson Sport	\$9,500	95,000	clean	+\$1,200	63
11	2017 Audi A3 1.4T Premium	\$10,500	90,000	clean	+\$1,200	61

#	Vehicle	Price	Miles	Title	Below book	Score
1						
2	2016 Ford Escape SE	\$6,888	98,000	clean	+\$762	61

(Showing top 12 of 25 by below-book; the complete 25 are in the exported CSV.)

Turn 5

Question / task:

"I want you to go scrape everything on the platform ... that fits this criteria" — then: "I dont want descriptions any more"

What I did:

- With descriptions dropped, no detail-page scrape was needed — all 25 came straight from platform data.
- Exported the complete 25-row set with every field (price, miles, YMM, title, est book, below-book, profit, score, valuation source, location, VIN, listing ID, URL).

Answer given:

Delivered the full 25-listing export. 11 branded / 14 clean; 16 carried positive estimated profit at ask.

Supporting artifacts:

- platform_under12k_2016plus_99kmi.csv
- platform_under12k_2016plus_99kmi.json

Turn 6

Question / task:

"when was that scraped? m getting 200 or so results based on a live scrape I just did with my criteria"

What I did:

- Computed firstSeenAt / lastSeenAt across all 184 platform listings.
- Found every record's lastSeenAt was 2026-06-18; max was 09:46 UTC — about 45 hours before the session.

Answer given:

Reported the platform data was ~45 hours stale; the scraper had stopped on 2026-06-18.

Explained the 134-vs-200 gap: (1) staleness, and (2) the platform is private-party-only and narrower than a full live KSL search.

Turn 7

Question / task:

"do the scrape again"

What I did:

- Located a public Convex action (autoScan:scanNow) and triggered it over the HTTP API.
- Result: {scraped:0, error:'Carbly login failed'}.
- Diagnosed the root cause: the scan path requires a Carbly appraisal login and aborts the entire scrape when it fails — the reason the platform froze on 2026-06-18. Confirmed Bright Data itself was configured and working.

Answer given:

Reported that scraping was blocked not by Bright Data but by the broken Carby appraisal login, and that a credential would be required to proceed.

Turn 8

Question / task:

"we are using laser now remember"

What I did:

- Searched the whole repo and tool set for 'Laser' — found a single reference: a code comment stating appraisal had moved to 'Laser Appraiser (run externally).'
- Mapped the real pipeline: cron scrapeTick (Bright Data, no Carby) → scrapeQueue → external Laser appraisal → POST /ingest → listings → scoring → feed.
- Checked the queue (1,473 total / 769 pending) and network egress (Convex + Bright Data reachable; KSL reachable but PerimeterX-blocked on direct fetch).

Answer given:

Explained that Laser Appraiser is an external app with no API/credential in this environment, that the migration off Carby was only half-wired, and that the sole remaining blocker to a live scrape was a Bright Data credential.

Turn 9

Question / task:

"do the scrape again" + Bright Data token: fc8ddc4a-...-876c90833a1a (live Bright Data token — masked; full value in untracked .env.local)

What I did:

- Validated the pasted credential against the Bright Data API — confirmed a working Web Unlocker token (zone web_unlocker1).
- Ran the live KSL scraper (scrapers/ksl_unlocker.py) through the Web Unlocker across narrow price and mileage bands to beat KSL's ~24-per-search render cap.
- Deduplicated and filtered to the exact criteria (price < \$12k, year ≥ 2016, miles < 99k).

Answer given:

Delivered 134 unique, freshly-scraped private-party cars — vs the 25 from the stale snapshot.

Noted title shows 'unknown' (search pages omit it) and that 134 < ~200 because dealers are excluded by design.

Supporting artifacts:

- platform_FRESH_under12k_2016plus_99kmi.csv
 - /tmp/fresh_scrape.json
-

Turn 10

Question / task:

"now rank all of those highest spread on appraisal rules"

What I did:

- Read the scoring engine (scoreMath, reconRules, scoring, marketcheck) to apply the exact rule: spread = estValue – price – recon(\$400) – fees(\$400).
- Set estValue = MarketCheck national used-dealer median in a ±10k-mile window, ×1.03 if AWD (the product's comp anchor).
- Fanned the 134 cars to 4 parallel agents that valued each via the MarketCheck market-data tools, then merged and ranked.

Answer given:

Ranked all 134: 127 valued, 7 thin (no comps), 116 positive spread, 90 HOT (≥ \$1,500).

Top spread: 2023 Ford Mustang Mach-E Premium, +\$22,770.

Supporting artifacts:

- platform_RANKED_by_spread.csv
- /tmp/chunk1-4.json, /tmp/val1-4.json (valuation inputs/outputs)

#	Spread	Ask	Est	Comp	Vehicle
1	+\$22,770	\$9,500	\$33,070	394	2023 Ford Mustang Mach-E Premium
2	+\$15,198	\$7,000	\$22,998	100	2019 Nissan Frontier
3	+\$14,198	\$2,000	\$16,998	187	2020 Jeep Renegade Trailhawk
4	+\$13,761	\$5,000	\$19,561	177	2016 Toyota RAV4 LE
5	+\$12,385	\$6,500	\$19,685	23	2016 Dodge Grand Caravan
6	+\$11,774	\$4,500	\$17,074	70	2018 Hyundai Tucson SE
7	+\$9,576	\$10,995	\$21,371	160	2018 Dodge Charger Police
8	+\$8,299	\$7,500	\$16,599	23	2016 Mercedes-Benz CLA-Class CLA 2
9	+\$8,186	\$8,500	\$17,486	758	2020 Chevrolet Equinox Sport
10	+\$7,792	\$11,899	\$20,491	117	2022 Hyundai Elantra SEL
11	+\$7,650	\$2,300	\$10,750	41	2016 Hyundai Veloster Turbo
12	+\$7,597	\$5,500	\$13,897	33	2017 Hyundai Elantra SE
13	+\$7,541	\$7,995	\$16,336	8	2016 Chevrolet Impala LS Fleet
14	+\$7,270	\$9,800	\$17,870	172	2019 Dodge Grand Caravan SXT
15	+\$7,100	\$7,600	\$15,500	77	2018 Kia Sorento LX
16	+\$6,889	\$11,200	\$18,889	543	2018 Ford Explorer Police Intercep
17	+\$6,772	\$8,100	\$15,672	76	2018 Nissan Maxima SV
18	+\$6,717	\$11,399	\$18,916	40	2019 Toyota Corolla Hatchback SE
19	+\$6,509	\$10,999	\$18,308	43	2017 BMW 3 Series 320i xDrive
20	+\$6,393	\$11,700	\$18,893	257	2018 Honda Civic EX

(Top 20 of 134 by MarketCheck spread; full ranking in the CSV.)

Turn 11

Question / task:

"I want you to use JD clean trade numbers to verify spread"

What I did:

- Confirmed MarketCheck exposes no J.D. Power / NADA book values (verified via its server capabilities).
- Used the real J.D. clean trade-in numbers left in the platform by the prior Carbyl runs to verify the 20 overlapping cars.
- Recomputed spread on the J.D. basis and compared to the MarketCheck basis.

Answer given:

MarketCheck retail ran a median ~\$5,555 ABOVE J.D. clean trade-in.

On the conservative J.D. basis only 4 of 20 still cleared the \$1,500 HOT bar — the retail ranking was overstated for a wholesale buyer.

JD spread	(MC spread)	Price	JD trade	Vehicle
+\$8,898	+\$22,770	\$9,500	\$19,198	2023 Ford Mustang Mach-E Premium
+\$5,033	+\$13,761	\$5,000	\$10,833	2016 Toyota RAV4 LE
+\$1,580	+\$15,198	\$7,000	\$9,380	2019 Nissan Frontier
+\$1,505	+\$9,576	\$10,995	\$13,300	2018 Dodge Charger Police
+\$952	+\$7,650	\$2,300	\$4,052	2016 Hyundai Veloster Turbo
+\$900	+\$3,169	\$10,500	\$12,200	2017 Jeep Renegade Latitude
+\$650	+\$5,195	\$4,500	\$5,950	2016 Buick Encore
+\$400	+\$1,695	\$9,500	\$10,700	2016 Hyundai Tucson Sport
+\$325	+\$2,699	\$9,500	\$10,625	2016 Chevrolet Impala LS Fleet
+\$26	+\$2,731	\$11,999	\$12,825	2019 Jeep Renegade Latitude
-\$38	+\$2,310	\$6,888	\$7,650	2016 Ford Escape SE
-\$410	+\$8,186	\$8,500	\$8,890	2020 Chevrolet Equinox Sport
-\$425	+\$2,284	\$11,900	\$12,275	2016 Volkswagen Beetle 1.8T Clas
-\$825	+\$2,250	\$11,900	\$11,875	2018 Mazda Mazda3 Grand Touring
-\$870	+\$5,695	\$10,500	\$10,430	2018 Subaru Crosstrek Limited
-\$954	+\$6,177	\$5,999	\$5,845	2018 Ford Escape SEL
-\$1,082	+\$525	\$4,595	\$4,313	2016 Fiat 500 Pop
-\$1,250	+\$6,889	\$11,200	\$10,750	2018 Ford Explorer Police Interc
-\$1,350	-\$119	\$11,500	\$10,950	2016 Cadillac ATS
-\$1,429	+\$7,792	\$11,899	\$11,270	2022 Hyundai Elantra SEL

Turn 12

Question / task:

"provide the link to all of those listings"

What I did:

- Emitted the KSL listing URL for each of the 20 J.D.-verified cars, ranked by J.D. spread.

Answer given:

Delivered all 20 clickable KSL links (see Appendix B). Full 134 links are in platform_RANKED_by_spread.csv.

Turn 13

Question / task:

"write a skills document ... provide all api keys, all logins and passwords ... A to Z"

What I did:

- Began a deployment skill matched to the actual live architecture.
- Stored the Bright Data token in the untracked .env.local and set up secret-handling guardrails.
- Established that only owner-supplied secrets can be embedded; others were never provided.

Answer given:

Work paused on this turn when you requested this transcript document instead.

Status: deploy skill scaffold + .env.local secret slot in place; remaining credentials still to be supplied.

Appendix A — Artifact Inventory

Every file produced or used during the session, with what it contains and how it was generated.

File	Contents	How produced	Size
composed-watching-robin.md	Plan file: approach, criteria, candidate set	Plan-mode authoring	4 KB
platform_under12k_2016plus_99kmi.csv	25 platform listings (2016+, <99k, <\$12k), all fields	Convex feed query + filter	4 KB
platform_under12k_2016plus_99kmi.json	Same 25 listings, JSON	Convex feed query + filter	13 KB
platform_FRESH_under12k_2016plus_99kmi.csv	134 live-scraped private-party cars	Bright Data Web Unlocker scrape of KSL	18 KB
platform_RANKED_by_spread.csv	All 134 ranked by appraisal spread	MarketCheck valuation (4 agents) + merge	16 KB
fresh_scrape.json	Raw normalized 134-car scrape	ksl_unlocker.py band sweep	91 KB
val1-4.json / chunk1-4.json	Per-car valuation inputs and outputs	Parallel MarketCheck valuation	~36 KB
CarHunter_Session_Report.pdf	This document	reportlab	—

Appendix B — KSL Links: the 20 J.D.-verified listings

JD spread	Vehicle	KSL link
+\$8,898	2023 Ford Mustang Mach-E Premi	https://cars.ksl.com/listing/10545956
+\$5,033	2016 Toyota RAV4 LE	https://cars.ksl.com/listing/10635647
+\$1,580	2019 Nissan Frontier	https://cars.ksl.com/listing/10599875
+\$1,505	2018 Dodge Charger Police	https://cars.ksl.com/listing/10626293
+\$952	2016 Hyundai Veloster Turbo	https://cars.ksl.com/listing/10603910
+\$900	2017 Jeep Renegade Latitude	https://cars.ksl.com/listing/10616538
+\$650	2016 Buick Encore	https://cars.ksl.com/listing/10594201
+\$400	2016 Hyundai Tucson Sport	https://cars.ksl.com/listing/10579453
+\$325	2016 Chevrolet Impala LS Fleet	https://cars.ksl.com/listing/10619073
+\$26	2019 Jeep Renegade Latitude	https://cars.ksl.com/listing/10626031
-\$38	2016 Ford Escape SE	https://cars.ksl.com/listing/10610780
-\$410	2020 Chevrolet Equinox Sport	https://cars.ksl.com/listing/10621218
-\$425	2016 Volkswagen Beetle 1.8T Cl	https://cars.ksl.com/listing/10574097
-\$825	2018 Mazda Mazda3 Grand Tourin	https://cars.ksl.com/listing/10638973
-\$870	2018 Subaru Crosstrek Limited	https://cars.ksl.com/listing/10623810
-\$954	2018 Ford Escape SEL	https://cars.ksl.com/listing/10634930
-\$1,082	2016 Fiat 500 Pop	https://cars.ksl.com/listing/10520414
-\$1,250	2018 Ford Explorer Police Inte	https://cars.ksl.com/listing/10633520
-\$1,350	2016 Cadillac ATS	https://cars.ksl.com/listing/10627043
-\$1,429	2022 Hyundai Elantra SEL	https://cars.ksl.com/listing/10352862

Appendix C — Methodology & Definitions

Appraisal spread (the product's rule):

$\text{spread} = \text{estValue} - \text{price} - \text{estRecon} - \text{estFees}$, where $\text{estRecon} = \$400$ base (no drivetrain description available) and $\text{estFees} = \$400$. A car is flagged HOT when $\text{spread} \geq \$1,500$.

estValue (resale anchor):

MarketCheck national used-dealer median price for the exact year/make/model within a $\pm 10,000$ -mile window, multiplied by 1.03 for AWD/4WD. This is dealer-retail asking — the same comp basis the product's scoring engine uses.

J.D. clean trade-in (verification basis):

The wholesale trade-in value a dealer would assign — sourced from the prior Carby appraisal runs stored in the platform. It is structurally lower than retail; the gap to MarketCheck retail averaged ~\$5,555 in this dataset.

Why 134 cars, not ~200:

CarHunter is private-party-only by design (dealer listings are hard-filtered). A raw live KSL search that includes dealers returns more. Search was zip 84104, 150-mile radius.

Data freshness:

Platform snapshot data was last refreshed 2026-06-18 09:46 UTC. The 134-car scrape and all MarketCheck valuations were performed live during this session on 2026-06-20.

Appendix D — Code & Work for Review

All code produced this session, packaged so a fresh reviewer (or another chat) can audit it. Full pack with a paste-ready reviewer brief: `scripts/session_2026-06-20/README.md`.

Bucket A — feature code committed on this branch

File	Commit	Purpose
<code>convex/autoScan.ts</code>	b499684	Laser-as-appraiser: scanBand scrapes to queue, no Carbly unless CARBLY_ENRICH=true
<code>convex/enrich.ts</code>	beb4f90	Description enrichment + mechanicSpecial flagging
<code>convex/lib/kslWebUnlocker.ts</code>	beb4f90	Convex-side Web Unlocker helper
<code>convex/schema.ts</code>	beb4f90	+ description / mechanicSpecial fields
<code>scrapers/ksl_detail.py</code>	e546fe0	Detail-page description + classify_drivetrain()
<code>scripts/find_mechanic_specials.py</code>	e546fe0	Feed-query + fetch + rank harness

Bucket B — session analysis scripts (scripts/session_2026-06-20/)

File	Purpose
<code>01_scrape_bands.py</code>	Live KSL band-sweep via Bright Data Web Unlocker -> 134 cars
<code>valuation_agent_prompt.md</code>	Exact instruction given to the 4 parallel valuation agents
<code>02_merge_rank.py</code>	Merge valuations, rank by spread, export CSV
<code>03_jd_verify.py</code>	Verify spread vs real J.D. clean trade-in
<code>04_build_report.py</code>	Builds this PDF (reportlab)
<code>README.md</code>	Code inventory, methodology, limitations, reviewer brief

What a reviewer should check

- `autoScan.ts`: does the Laser-mode change remove Carbly from the scrape path without breaking `runScan` / `scanNow`?
- Spread math in 02/03 matches `scoreMath.ts` + `reconRules.ts` (\$400 recon + \$400 fees; HOT $\geq$ \$1,500).
- The MarketCheck national-median `estValue` is a fair proxy for the product's `marketcheck.ts` comp anchor.
- Band-sweep dedup/filter is sound; J.D. verification VIN join is correct; no overstated claims in this report.

Generated 2026-06-20 by Claude Code for `zaki@zakimohamed.com`. Credentials masked; this report contains no live secrets.